

Dictionaries, Keys, Values, JSON, and Github

Introduction

In this assignment, we must create a program that can register a student for a course. The user has to be able to input a first and last name, in addition to a course name. Unlike the previous assignment, dicts are used in place of lists. We will also learn how to upload code to github.

Understanding & Importing JSONs

JSON stands for Javascript Object Notation. It allows us to store many types of data and nest them, if needed. It differs from the CSVs we have been using because while its file size is larger, it can store hierarchies of complex data. An example would be using JSONs to store enemy data in a video game (integer for health, and strings for names). Then using a CSV to hold user account information like emails and usernames, since this dataset would most certainly be bigger. To import JSON into PyCharm, we simply type figure 1.

Figure 1: Importing JSONs.

```
import json
```

Implementing Dictionaries

We use dictionaries (dicts) to store data. The “key” refers to the column, and the value is what is stored inside. Name would be a key, and each person’s name is the value. We use {} brackets to define dicts and refer to them in code. This is in contrast to lists, which use normal brackets [].

Setting Up

To prepare our program, we have to import JSON and present a menu string. We also have to set our FILE_NAME to our existing JSON file for enrollment data. Then, we set

variables like: names, rows, and our table. We use the function load to read the file data into our table as shown in figure 2.

Figure 2: Setting variables, constants, and more.

```
import json

# Define the Data Constants
MENU: str = '''
---- Course Registration Program ----
Select from the following menu:
    1. Register a Student for a Course.
    2. Show current data.
    3. Save data to a file.
    4. Exit the program.
-----
'''

# Define the Data Constants
FILE_NAME: str = "Enrollments.json"

# Define the Data Variables and constants
student_first_name: str = '' # Holds the first name of a student entered by
the user.
student_last_name: str = '' # Holds the last name of a student entered by the
user.
course_name: str = '' # Holds the name of a course entered by the user.
student_data: dict = {} # one row of student data
students: list = [] # a table of student data
file = None # Holds a reference to an opened file.
menu_choice: str = "" # Hold the choice made by the user.

# When the program starts, read the file data into a list of lists (table)
# Extract the data from the file
file = open(FILE_NAME, "r")
data = json.load(file)
file.close()

# Check if a file object exists and is still open
if file is not None and file.closed == False:
    file.close()

# Present the menu of choices
print(MENU)
menu_choice = input("What would you like to do: ")
```

Implementing Menu Choices

In the first menu choice, we use the append function to add our row (student_data) to our table students. Line 61 demonstrates that to refer to dicts, we must use {} brackets instead of [].

Figure 3: Menu choice 1.

```
# Input user data
if menu_choice == "1": # This will not work if it is an integer!
    student_first_name = input("Enter the student's first name: ")
    student_last_name = input("Enter the student's last name: ")
    course_name = input("Please enter the name of the course: ")

    student_data = {"student_first_name":
student_first_name,"student_last_name": student_last_name,"course_name":
course_name}
    students.append(student_data)
    print(f"You have registered {student_first_name} {student_last_name} for
{course_name}.")
    continue
```

For the second menu choice, we again must use {} brackets to refer to keys in the dict. For each student in our table of students, we tell the program to print one out with a message for readability. This is more user friendly than simplifying displaying a table.

Figure 4: Menu choice 2.

```
# Present the current data
elif menu_choice == "2":
    # Process the data to create and display a custom message
    #added for __ in __
    print("-" * 50)
    for student in students:
        print(
            f"Student {student['student_first_name']}
{student['student_last_name']} is enrolled in {student['course_name']}")
    print("-" * 50)
    continue
```

Finally, for the third and fourth choice, we allow the data entered to be saved. We use the dump function to write our data into the file. We display a message to the user letting them know what has been saved. Then, in the fourth choice, the user is able to break out of the loop and exit the program.

Figure 5: Menu choices 3 and 4.

```
# Save the data to a file
elif menu_choice == "3":
```

```

    file = open(FILE_NAME, "w")
    json.dump(students, file, indent=4)
    file.close()

    # Check if a file object exists and is still open
    if file is not None and file.closed == False:
        file.close()

    print("The following data was saved to file!")
    for student in students:
        print(f"Student {student['student_first_name']}
{student['student_last_name']} is enrolled in {student['course_name']}")
        continue

    # Stop the loop
    elif menu_choice == "4":
        break # out of the loop
    else:
        print("Please only choose option 1, 2, 3, or 4")

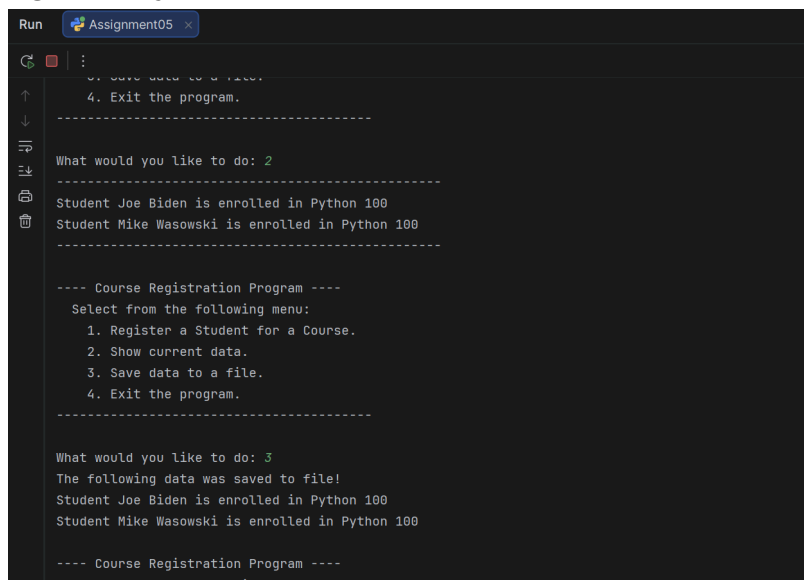
print("Program Ended")

```

Testing the Final Code

The script works in PyCharm and the command console.

Figure 6: PyCharm execution successful.



The screenshot shows the PyCharm Run window for a file named 'Assignment05'. The output of the program is displayed in a dark-themed console. The program starts with a menu: '1. Register a Student for a Course.', '2. Show current data.', '3. Save data to a file.', and '4. Exit the program.' The user selects option 2, and the program displays the current data: 'Student Joe Biden is enrolled in Python 100' and 'Student Mike Wasowski is enrolled in Python 100'. The user then selects option 3, and the program displays the message 'The following data was saved to file!' followed by the same student data. The program then returns to the menu.

```

Run Assignment05 x
4. Exit the program.
-----
What would you like to do: 2
-----
Student Joe Biden is enrolled in Python 100
Student Mike Wasowski is enrolled in Python 100
-----

---- Course Registration Program ----
Select from the following menu:
1. Register a Student for a Course.
2. Show current data.
3. Save data to a file.
4. Exit the program.
-----

What would you like to do: 3
The following data was saved to file!
Student Joe Biden is enrolled in Python 100
Student Mike Wasowski is enrolled in Python 100

---- Course Registration Program ----
Select from the following menu:

```

Figure 7: Command prompt execution successful.

```

C:\Users\hidde\Documents\Python\PythonCourse\_Module05\A05>python Assignment05.py

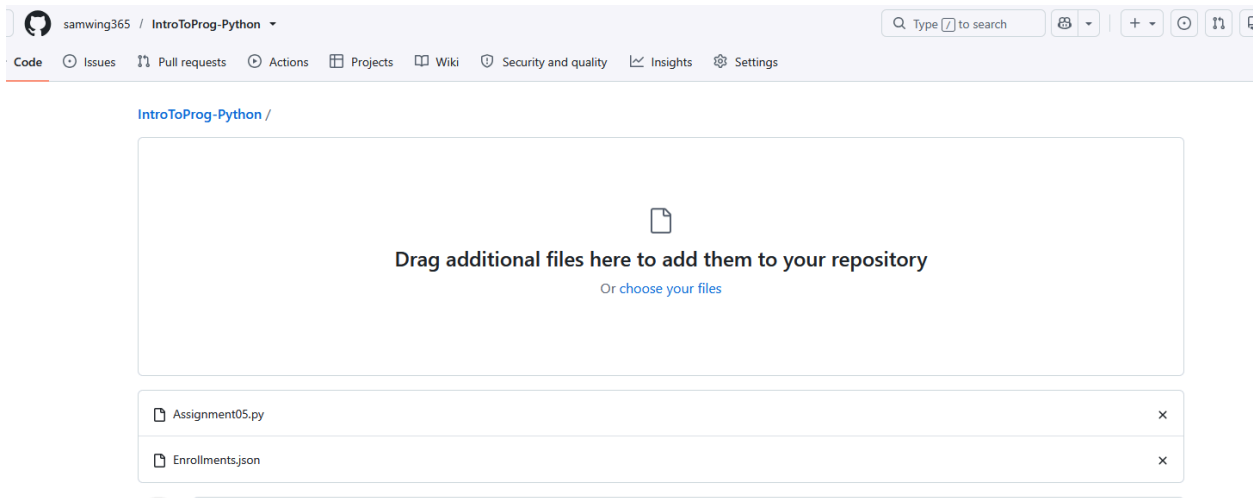
---- Course Registration Program ----
Select from the following menu:
  1. Register a Student for a Course.
  2. Show current data.
  3. Save data to a file.
  4. Exit the program.
-----

What would you like to do: 1
Enter the student's first name: Joe
Enter the student's last name: Wheeler

```

Using Github

Github is a service which allows its users to share code easily using repositories. We are able to upload many types of files and get feedback or edit code with others. Figure 8 demonstrates this.



Summary

In this assignment, we were introduced to dicts, JSON, and Github. Dicts were used in this assignment to hold rows of student data about their name and course. A JSON was used to store this data for multiple students. Github was used to upload the code.